

Multi-Stage Transfer Learning for Time-Frequency Sound Event Detection: From Images to Bioacoustics to Industrial Audio

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Abstract—Sound event detection (SED) has mainly focused on the temporal extent of acoustic events, but certain applications require the detection of their time-frequency bounding boxes (BBs), which is referred to as time-frequency SED (TF-SED). In industrial audio, detecting BBs of whistle-like sounds caused by defects on a wind turbine blade is valuable because their shape in the spectrogram carries information on the damage severity. However, TF-SED remains underexplored partly because annotated datasets are scarce owing to the expertise required for annotation in bioacoustics and data confidentiality in industry. Prior work addressed TF-SED from a model-design perspective by combining an audio spectrogram transformer backbone with a detection transformer detector, while the training strategy was left unexplored. This study investigates a multi-stage transfer learning strategy that bridges modality and domain gaps and trains the model in three steps: training on image object detection data (COCO), bird audio data (BirdSet), and wind turbine audio data (WindTurbine). In the final stage, we apply component-wise progressive unfreezing, gradually expanding the set of learnable components. Our experiments revealed three findings. First, transfer learning from COCO is beneficial despite the modality gap, as the benefit of large-scale pretraining outweighs the cost of switching modality. Second, when transferring through the intermediate audio domain, keeping the backbone frozen, thereby preserving the general-purpose representation from large-scale pretraining and not specializing in the intermediate domain, yields better generalization to the target domain. Third, in the final adaptation stage, unfreezing the object query before the rest of the detector is effective because it encodes the characteristics of the target sound events. These findings provide practical guidance for efficiently training TF-SED models in other target domains using limited annotated data.

Index Terms—Time-frequency sound event detection, multi-stage transfer learning, progressive unfreezing, wind turbine monitoring

1. INTRODUCTION

Sound event detection (SED) has primarily focused on determining the presence and temporal extent (onset and offset times) of acoustic events. However, certain applications require localizing events in both time and frequency, that is, detecting a two-dimensional time-frequency bounding box (BB) for each event. This task is referred to as time-frequency SED (TF-SED), in contrast to conventional temporal SED. By treating a spectrogram as an image-like representation, TF-SED can be naturally framed as an audio counterpart of object detection.

In bioacoustics, bird vocalization detection illustrates this need: in complex acoustic environments where vocalizations of multiple individuals overlap, time-frequency BBs enable the localization of each vocalization by species and individual [1], [2]. Furthermore, BBs correspond to acoustic units, and understanding how such units form higher-level vocal structures remains a challenging research problem [3]. Audio recordings annotated with time-frequency BBs by domain experts are referred to as fully annotated [1], [4].

In industrial audio, early detection of blade damage (e.g., cracks or holes) in wind turbines is critical for maintenance. If left unaddressed, such damage not only reduces power generation efficiency and increases noise but can also lead to blade fracture or detachment, posing serious safety risks. As the blade rotates, these defects produce a whistle-like sound. Detecting these acoustic events enables early identification of damage. This monitoring approach requires only

commodity microphones attached to the turbine tower and enables continuous, nonintrusive monitoring. Importantly, the time-frequency shape of these anomalous sounds carries information relevant to estimating the severity of the underlying damage, although estimating the damage severity from the detected shape is beyond the scope of this study and is left for future work. An SED system that identifies only the temporal interval of an event cannot inform decisions regarding whether a detailed inspection or repair is warranted. The detection of sound events via time-frequency BBs is therefore essential.

Despite its importance, TF-SED remains underexplored partly because annotated datasets are scarce. In bioacoustics, large-scale audio data are publicly available, particularly for bird vocalization. However, the time-frequency BB annotation requires domain expertise, making it costly to construct labels for new domains. In an industrial setting, data confidentiality limits the availability of public datasets.

A previous study [5] addressed wind turbine TF-SED from a model design perspective and demonstrated that an architecture combining an audio spectrogram transformer (AST) backbone, which extracts an embedding representation from the input spectrogram, with a detection transformer (DETR) substantially outperforms conventional approaches. However, the training strategy, specifically how to make the most of related out-of-domain data given the scarcity of in-domain annotations, has not yet been investigated. While transfer learning for object detection has been studied across diverse target domains, including space imagery [6], remote sensing [7], and medical imaging [8], transfer learning strategies spanning both the modality and domain gaps for TF-SED remain unexplored.

This study systematically investigates multi-stage transfer learning across modality and domain gaps for TF-SED, deriving practical design principles for such strategies. Specifically, we reveal the following insights. (1) First, transferring from a large-scale image object detection dataset (COCO) is beneficial despite the modality gap, as the benefit of large-scale pretraining outweighs the cost of switching modalities. (2) Second, when transferring through an intermediate audio source domain (bird vocalization detection on BirdSet), keeping the backbone frozen, thereby preserving the general-purpose representation learned from large-scale AudioSet pretraining, yields better generalization to the final target domain than allowing it to specialize in the intermediate domain. (3) Third, during the final adaptation stage in the target domain (wind turbine sounds), unfreezing the object query before the rest of the detector is effective, while a separate stage to unfreeze the detector before the backbone is not necessarily beneficial and can be detrimental. These findings provide practical guidance for the efficient training of TF-SED models in other target domains using limited annotated data.

The remainder of this paper is organized as follows. First, we describe the model architecture (Section 2) and datasets (Section 3), both of which largely follow the previous study [5]. Then, we present our main contribution, a multi-stage transfer learning strategy, in Section 4. Experiments and results are presented in Section 5, followed by the concluding remarks in Section 6.

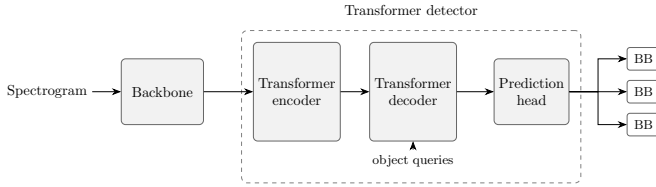


Fig. 1: Model architecture. It consists of a backbone and a transformer detector, which comprises a detection encoder-decoder and a prediction head. Object queries are fed into the transformer decoder.

2. MODEL

We adopt the backbone-detector architecture proposed in the previous study [5] for wind turbine TF-SED and keep this architecture throughout this study. This study focuses on a transfer learning strategy to train the model (Section 4). The architecture follows the detection transformer (DETR) formulation [9], which is decomposed into two parts: a backbone that extracts an embedding representation from the input spectrogram and a transformer detector that predicts a set of BBs from these representations. The transformer detector consists of three components: a set of object queries representing candidate detections; a detection encoder-decoder that refines these queries using the backbone embedding; and a prediction head that maps the refined queries to final box coordinates and scores. We employ this terminology throughout the remainder of this paper. The architecture is illustrated in Fig. 1.

The backbone is the efficient audio transformer (EAT) [10], a variant of audio spectrogram transformers (ASTs) [11] pretrained on AudioSet [12], a large-scale dataset of environmental sounds. ASTs share the same transformer-based architecture as a vision transformer (ViT) [13], which is applied to spectrogram inputs instead of images. EAT was selected for its strong benchmark performance on audio classification tasks among AST variants. We retain this choice and leave the comparison of different AST backbones for TF-SED to future work, as this study focuses on the transfer learning strategy rather than the choice of backbone architecture.

As the transformer detector, DINO (DETR with improved denoising anchor boxes) [14] is used. To capture features at multiple scales, a simple feature pyramid (SFP) [15] is inserted as the neck immediately after the backbone, converting its output embedding into a set of multi-scale feature maps.

This combination of the backbone, neck, and transformer detector has been shown to outperform the conventional approach using a region-based convolutional neural network with a residual network backbone for wind turbine TF-SED [5]. Hence, we keep this architecture unchanged in the present study.

3. DATA

For the image source domain, we use the common objects in the context (COCO) dataset [16], which is a large-scale dataset for object detection in natural images.

As an audio source domain, we use BirdSet [1], which is a large-scale dataset for bird vocalization detection recorded at a sampling rate of 32 kHz. The training and test sets differ in annotation granularity; the training set is weakly annotated with onset and offset times, whereas the test set is fully annotated with time-frequency BBs. We segmented all test-split recordings into 15-s segments, yielding 44,435 segments (185 h in total) with 149,884 boxes (3.4 per segment) across 454 species classes.

As only the test set contains time-frequency BB annotations, it is used for the intermediate transfer learning step (Step 2 in Section 4),

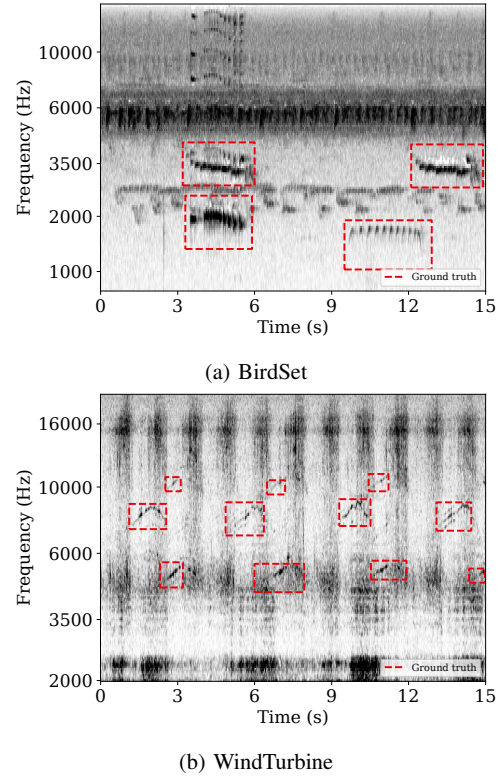


Fig. 2: Example spectrograms with time-frequency BB annotations.

whereas the training set is not used in this study. Leveraging the weakly labeled data is left for future research. As the objective is to transfer general time-frequency detection ability to the target domain, bird vocalization detection performance is not evaluated. The original species labels are disregarded and all annotated events are treated as a single class. Fig. 2a shows an example spectrogram with time-frequency BB annotations.

As the target domain, we use the wind turbine sound dataset collected in the previous study [5], which we refer to as the WindTurbine dataset in this paper. It consists of recordings of sounds produced by land-based wind turbines at two locations over one year at a sampling rate of 48 kHz. A subset of the recordings is annotated with time-frequency BBs, comprising 1,843 segments of 15 s each (7.6 h in total) and 14,420 boxes (7.8 per segment), with frequencies ranging from 2 to 20 kHz. As no taxonomy of whistle-like damage sounds has been established, all annotated events belong to a single class. Fig. 2b shows an example spectrogram with the time-frequency BB annotations.

Note that BirdSet and WindTurbine differ in their sampling rates (32 kHz vs. 48 kHz). We treat this as a nonissue because detecting BBs does not require preserving the physical frequency scale of the spectrogram; therefore, we retain both at their original sampling rate. We avoid downsampling WindTurbine to BirdSet’s rate because this would discard its highest-frequency boxes, and upsampling BirdSet offers no clear benefit as confirmed in our preliminary experiments.

4. METHOD

We train the model through three steps of transfer learning, progressing from the image source domain to the audio source domain and finally to the target domain: Step 1 on COCO (image), Step 2 on BirdSet (bioacoustics), and Step 3 on WindTurbine (industrial audio). We refer to this three-step procedure as multi-stage transfer learning. Fig. 3

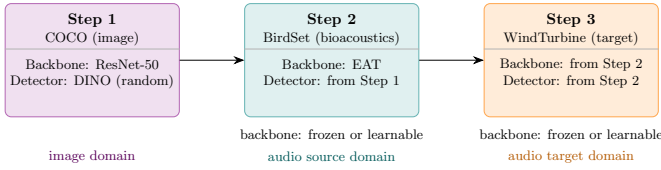


Fig. 3: Overview of the three-step multi-stage transfer learning procedure, progressing from the image domain (COCO) to the audio source domain (BirdSet) to the audio target domain (WindTurbine).

summarizes the backbone, transformer detector, and dataset used at each step, alongside the components initialized from the previous step versus the newly introduced components. The final adaptation stage in the target domain, Step 3, additionally incorporates a component-wise progressive unfreezing procedure for the backbone and components of the transformer detector, as described at the end of this section.

In Step 1, we use the publicly available DINO model [14] pretrained on COCO [16] with a ResNet-50 [17] backbone pretrained on ImageNet [18]. This provides the transformer detector with a starting point learned from image object detection before it is exposed to audio data. Note that the SFP neck described in Section 2 is designed for the AST-based backbone used in Step 2 onward, and is thus not used in Step 1, where the backbone is ResNet-50. We evaluate the extent to which initializing the transformer detector from this state, pretrained on COCO, improves the final target-domain performance compared to random initialization.

In Step 2, we transfer the model trained in Step 1 to BirdSet, replacing the ResNet-50 backbone with EAT, while reusing the transformer detector trained in Step 1. The SFP neck, as described in Section 2, is introduced following prior studies showing that multi-scale feature maps are required when connecting a ViT-style backbone to a DETR-family detector [5], [15]. The asymmetric treatment of these two parts reflects their different roles. As the input modality changes from natural images to spectrograms, the backbone is reinitialized using audio-domain pretrained weights (AudioSet), rather than inheriting the image-domain weights from Step 1. By contrast, the transformer detector, which localizes events from embeddings independent of the input modality, is expected to transfer across this gap and is thus carried over from Step 1. In this step, we compare two conditions for the backbone: keeping it frozen at AudioSet-pretrained weights versus allowing it to adapt to BirdSet.

In Step 3, we transfer the model to WindTurbine, which is the target domain. At this final stage, we additionally apply a component-wise progressive unfreezing strategy across the backbone and transformer detector, unfreezing their components incrementally rather than simultaneously. Prior work explored freezing and unfreezing strategies over components of DETR models [19]–[22]. One approach kept the entire detector frozen and augmented object queries with external prompts [23]. However, training the object query preferentially has not been investigated, although it is expected to encode the shape and characteristics of the target acoustic events. The object query is therefore trained first in this study. The prediction head remains trainable throughout all three configurations described below, as it directly determines the final output and cannot be fixed when adapting to a new domain. We examine whether unfreezing the transformer detector prior to the backbone provides any additional benefit compared with jointly unfreezing all components. In either case, the backbone is unfrozen in the final step. Fig. 4 summarizes the three resulting configurations (3-a, 3-b, and 3-c) and indicates which components are frozen or learnable.

Component	3-a	3-b	3-c
Backbone	frozen	frozen	learnable
Object query	learnable	learnable	learnable
Encoder-decoder	frozen	learnable	learnable
Prediction head	learnable	learnable	learnable

Fig. 4: Component-wise progressive unfreezing schedule used in Step 3. Backbone is unfrozen only in the final configuration (3-c), regardless of the unfreezing strategy for the transformer detector.

5. EXPERIMENTS

5.1. Setup

The model architecture follows that of the prior study [5] without modification. The model was trained using the AdamW optimizer with a weight decay of 1×10^{-4} . The learning rate was set to 1×10^{-4} for the transformer detector, and 1×10^{-5} for the EAT backbone. Gradient clipping with maximum L2 norm of 0.1 was applied. The loss function follows the original DINO formulation [14] and comprises a Hungarian matching loss, classification loss, and box regression loss, consisting of an L1 loss and a generalized intersection over union (IoU) loss. Hyperparameters were adopted from a previous study [14] without additional tuning. In each step, the model was trained for up to 12 epochs, with the best epoch selected based on the validation performance. In the progressive unfreezing stage, Steps 3-a and 3-b each lasted 2 epochs, followed by Step 3-c for 12 epochs.

For BirdSet, the test split provided by the dataset was divided into training, validation, and test sets at a ratio of 7:1:2 at the segment level. For WindTurbine, the recordings were divided at the same ratio (7:1:2) at the segment level, ensuring that all recordings from the same date and location were assigned to the same set. The spectrograms were computed using a mel filterbank with a window size of 25 ms and a hop size of 10 ms. The input to the model consists of 15-s segments (1,498 frames) with 182 mel frequency bins. The frequency range was set to 20 Hz–16 kHz for BirdSet and 2–20 kHz for WindTurbine.

We evaluated the detection performance using the average precision at an IoU threshold of 50% (AP_{50}), following the COCO evaluation protocol [16]. As noted in prior work [5], the boundaries of whistle-like sound events in spectrograms are inherently ambiguous compared with objects in natural images. Therefore, an IoU threshold of 50% is more appropriate than stricter thresholds commonly used in image object detection.

5.2. Results

Table 1 shows the AP_{50} values of the wind turbine TF-SED under different multi-stage transfer-learning conditions. First, we consider the effect of transfer from the image source domain. Training on WindTurbine alone yielded an AP_{50} of 0.392, whereas initializing the transformer detector from a state pretrained on COCO (Step 1 $\rightarrow$ 3) substantially improved the performance to 0.479. This demonstrates that image-domain pretraining is beneficial for TF-SED despite the modality gap, indicating that the benefit of large-scale pretraining outweighs the cost of switching modalities.

Next, we examine the effect of introducing an intermediate audio source domain (Step 2) and the effect of freezing the backbone at this stage. Comparing Step 1 $\rightarrow$ 2 $\rightarrow$ 3 with Step 1 $\rightarrow$ 3, adding a BirdSet intermediate step with a frozen backbone (0.484) outperformed skipping it (0.479). This result confirms the benefit of transferring

Table 1: Performance of wind turbine TF-SED (AP_{50}) under different multi-stage transfer learning conditions. The “progressive” column indicates whether component-wise progressive unfreezing is applied in Step 3. The proposed method (bottom row) employs the best progressive unfreezing strategy.

Transfer learning	Step 1 (COCO) Training	Step 2 (BirdSet)		Step 3 (WindTurbine)		
		Training	Backbone	Training	Progressive	Backbone frozen learnable
3			-	✓		0.385 0.392
1 → 3	✓		-	✓		0.409 0.479
2 → 3		✓	learnable	✓		0.371 0.443
2 → 3		✓	frozen	✓		0.424 0.468
1 → 2 → 3	✓	✓	learnable	✓		0.428 0.478
1 → 2 → 3	✓	✓	frozen	✓		0.434 0.484
1 → 2 → 3 (proposed)	✓	✓	frozen	✓	✓	0.435 0.504

Table 2: Ablation study of component-wise progressive unfreezing in Step 3 using the proposed multi-stage transfer learning condition (Step 1 → 2 (backbone frozen) → 3).

Progressive unfreezing	AP_{50}
3-c	0.484
3-a → 3-c	0.504
3-b → 3-c	0.485
3-a → 3-b → 3-c	0.477
3-b	0.434
3-a → 3-b	0.435

through the intermediate audio source domain. Furthermore, keeping the backbone frozen during Step 2 consistently outperformed allowing it to adapt: 0.468 vs. 0.443 for Step 2 → 3 and 0.484 vs. 0.478 for Step 1 → 2 → 3. This suggests that preserving the general-purpose representation acquired through AudioSet pretraining, rather than specializing the backbone in the intermediate domain, leads to better generalization to the target domain.

Table 2 presents an ablation study of component-wise progressive unfreezing in Step 3 based on the best multi-stage transfer learning condition (Step 1 → 2 (backbone frozen) → 3). Progressive unfreezing with the sequence 3-a → 3-c achieved the best performance (0.504), outperforming the baseline without progressive unfreezing (3-c: 0.484). This confirms that unfreezing the object query before the rest of the detector is effective, as it allows the model to first adapt the component that encodes the characteristics of the target acoustic events. Among the other sequences, 3-b → 3-c (0.485) was comparable to 3-c alone. This suggests that unfreezing the detection encoder-decoder prior to the backbone is not necessarily beneficial. Interestingly, 3-a → 3-b → 3-c (0.477) underperformed 3-a → 3-c (0.504), despite the fact that 3-a → 3-b (0.435) performed comparably to 3-b alone (0.434). Based on this observation, Step 3-b may subtly corrupt the internal representation acquired in Step 3-a in a manner that is not immediately apparent from the intermediate performance, but manifests when the backbone is jointly fine-tuned in Step 3-c. We hypothesize that unfreezing the encoder-decoder while the backbone is still frozen (Step 3-b) causes it to overfit to the intermediate backbone representations. When the backbone is subsequently unfrozen (Step 3-c), the updated backbone representations cause a severe mismatch, degrading the detection performance. Verifying this hypothesis is left for future work.

Finally, making the backbone learnable in Step 3 consistently improved the performance across all conditions in Table 1, regardless of the multi-stage transfer learning strategy employed. This indicates that fine-tuning the backbone in the target domain is always beneficial, even when the dataset is limited in size.

6. CONCLUSION

This study investigated a multi-stage transfer learning strategy for time-frequency SED (TF-SED) spanning modality and domain gaps, with wind turbine anomaly detection as the target application. In the proposed method, training proceeds in three steps: training on image object detection data (COCO), bird audio data (BirdSet) as an intermediate audio source domain, and wind turbine audio data (WindTurbine) as the target domain. Within the final stage, component-wise progressive unfreezing was applied. Our experiments revealed three findings: (1) transfer learning from a large-scale image dataset is beneficial despite the modality gap; (2) keeping the backbone frozen during the intermediate audio domain stage, thereby preserving the general-purpose representation from large-scale pretraining, yields better generalization to the target domain; and (3) unfreezing the object query before the rest of the detector is effective during the final adaptation stage, while introducing a separate stage to unfreeze the detector prior to the backbone can be detrimental. These findings provide practical design principles for efficiently training TF-SED models in data-scarce target domains.

This study suggests several directions for future research. The proposed framework can be extended in terms of data utilization and domain coverage. In TF-SED, fully annotated data with time-frequency BBs are scarce across domains; leveraging unannotated or weakly annotated data through semi-supervised learning thus represents a promising direction, drawing on prior work in object detection [24]–[26]. The transfer learning framework explored here can be applied in the reverse direction, from industrial audio to bioacoustics, and extended within bioacoustics using BirdSet as a source domain for other species, including frogs, bees, whales, and bats. From a methodological perspective, replacing ResNet-50 with a ViT-style backbone in Step 1 and systematically exploring progressive unfreezing strategies, potentially through automated search, may yield further performance gains and deeper insights into component-wise adaptation dynamics. In addition, parameter-efficient fine-tuning methods such as low-rank adaptation (LoRA), which have recently been applied to DETR models [27], [28], represent a promising direction for reducing the annotation burden associated with adapting to new target domains.

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